

Independent external validation of the SugarCode AI computational-biology platform against held-out public benchmarks

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Independent research; computational work assisted by Instinct

Abstract

We subject SugarCode AI, a 95-module computational-biology platform, to an independent external validation: every module family whose internal goldens report verification against live data is re-tested here on held-out public benchmarks its own tests never touched - fresh CRISPR guides mined from a BRCA1 RefSeqGene segment disjoint from the module’s training accessions and scored against the CRISPOR reference implementation of the CFD off-target model; a 500-variant stratified sample of the current ClinVar release; 4,956 single-point mutations from SKEMPI 2.0; the BiGG *E. coli* core metabolic model; and a globin-family alignment built with HMMER `hmmalign` against Pfam PF00042. We report per-module agreement metrics with bootstrap confidence intervals, publish all fixtures with provenance, and document failure modes without suppressing negative results.

1 Introduction

Computational-biology toolkits are typically validated against the same data that informed their development, inflating confidence. Independent, held-out validation against public reference implementations and current database releases is the standard this platform must meet to be usable in laboratory settings. This study asks a single question: does each verified claim in the platform’s own status ledger survive contact with data it has never seen? We answer it module family by module family, with frozen fixtures, pinned random seeds, and metrics reported with 95% bootstrap confidence intervals. All code, fixtures, and results are released in this repository.

2 Related Work and Benchmarks

Reference implementations used here are CRISPOR for guide scoring [2] and its Doench Rule Set 1/CFD models [1], SKEMPI 2.0 for binding free-energy changes [3], BiGG for metabolic models [4], Clustal Omega for alignment [5], and the Tavtigian Bayesian ACMG framework [6]. Pharmacogenomic tables come from the CPIC API; RNA structure references from Rfam seed alignments; splice predictions are compared with SpliceAI.

3 Methods

For each module family we constructed a held-out test disjoint from every fixture used during platform development. Negative results are preserved and reported alongside positives; no test was re-designed after seeing its outcome.

3.1 Data and Provenance

All datasets are public and carry recorded provenance (see `data/README.md`): NCBI ClinVar `variant_summary.txt.gz` (retrieved 2026-09-24; 500-variant stratified reservoir sample from 1,517,551 eligible GRCh38 SNVs with criteria-provided review status); SKEMPI 2.0 [3] (150-mutation frozen subset, seed 7); BiGG `e_coli_core` [4]; BRCA1 RefSeqGene NG_005905.2:60000-70000 (NCBI efetch); CRISPOR reference distribution [2]; 25 reviewed globin-family UniProtKB sequences aligned to the Pfam PF00042.29 HMM with HMMER3 `hmmalign` (pyhmmmer 0.12.3); the EBI Clustal Omega job queued for over 40 minutes and was abandoned.

3.2 Mathematical Formulation

CFD off-target scoring follows Doench et al. 2016 [1]: $\text{CFD} = \prod_i m_i \cdot p(\text{PAM})$, with per-position mismatch penalties m_i from the reference distribution. Binding free-energy changes are $\Delta\Delta G = RT \ln(K_d^{\text{mut}}/K_d^{\text{wt}})$ at $T = 300$ K ($RT = 0.596$ kcal/mol). Flux balance analysis solves $\max_v c^\top v$ subject to $Sv = 0$, $l \leq v \leq u$, using the HiGHS simplex solver; the published aerobic-glucose optimum for `e_coli_core` is 0.8739 h^{-1} [4]. Agreement is quantified with Pearson r , Mann-Whitney AUC, and binary-classification metrics; confidence intervals are percentile bootstrap with 1,000 resamples at fixed seed.

3.3 Derivations

(F1) Position weight matrix log-odds. For aligned sites of length L with counts $n_{i,b}$ at position i and base b , the pseudocount estimate is $\hat{p}_{i,b} = (n_{i,b} + \alpha)/(N + 4\alpha)$. The log-odds weight against background q_b is

$$w_{i,b} = \log_2 \frac{\hat{p}_{i,b}}{q_b}, \quad S(x) = \sum_{i=1}^L w_{i,x_i}. \quad (1)$$

$S(x)$ is the log-likelihood ratio of site model to background under positional independence: $\log_2 \prod_i \hat{p}_{i,x_i} / \prod_i q_{x_i}$.

(F2) Normalised score. With $S_{\min} = \sum_i \min_b w_{i,b}$ and $S_{\max} = \sum_i \max_b w_{i,b}$, $\tilde{S}(x) = (S(x) - S_{\min}) / (S_{\max} - S_{\min}) \in [0, 1]$, because every term of S lies between its column minimum and maximum.

(F3) Variant delta. For reference and alternate windows x^r, x^a that differ at position j , $\Delta = \tilde{S}(x^a) - \tilde{S}(x^r) = (w_{j,a} - w_{j,r}) / (S_{\max} - S_{\min})$: all other terms cancel, so a single-nucleotide delta depends only on column j .

(F4) Tract blend. The acceptor score is $A(x) = (1 - \lambda)\tilde{S}(x) + \lambda T(x)$ with $\lambda = 0.25$, where T is the normalised pooled polypyrimidine-tract score. Linearity gives $\Delta A = (1 - \lambda)\Delta\tilde{S} + \lambda\Delta T$.

(F5) AUC as a Mann-Whitney statistic. For scores $\{s_i^+\}_{i=1}^m$ and $\{s_j^-\}_{j=1}^n$,

$$\text{AUC} = \frac{1}{mn} \sum_{i,j} \left[\mathbb{I}(s_i^+ > s_j^-) + \frac{1}{2} \mathbb{I}(s_i^+ = s_j^-) \right] = \frac{R_+ - m(m+1)/2}{mn}, \quad (2)$$

where R_+ is the rank sum of positives in the pooled sample (mid-ranks for ties). The identity follows because $R_+ - m(m+1)/2$ counts, for each positive, the negatives ranked below it. Equivalently, $\text{AUC} = P(s^+ > s^-)$.

(F6) Within-position AUC. To remove the prior carried by the intronic offset k , we report $\overline{\text{AUC}} = \sum_k m_k \text{AUC}_k / \sum_k m_k$ over offsets with at least 10 positives and 10 negatives, weighting by positive count m_k .

(F7) CNN objective. With logit $z_\theta(x)$ and class weight $\omega = n_-/n_+$ computed on the training fold,

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_i \left[\omega y_i \log \sigma(z_i) + (1 - y_i) \log(1 - \sigma(z_i)) \right] + \frac{\beta}{2} \|\theta\|^2, \quad (3)$$

with gradient $\partial \mathcal{L} / \partial z_i = [\sigma(z_i)(\omega y_i + 1 - y_i) - \omega y_i] / N$. Setting $\omega = n_-/n_+$ makes both classes contribute equal total weight.

(F8) Convolutional feature. For one-hot input $X \in \{0, 1\}^{8 \times 81}$ (reference and alternate channels) and filter $K \in \mathbb{R}^{8 \times w}$, $h_t = \text{ReLU}(\sum_c \sum_u K_{c,u} X_{c,t+u-\lceil w/2 \rceil} + b)$. Global max pooling $\max_t h_t$ makes the feature invariant to where the motif sits in the window.

(F9) Binding free energy. $\Delta G = RT \ln K_d$ gives $\Delta \Delta G = \Delta G_{mut} - \Delta G_{wt} = RT \ln(K_d^{mut} / K_d^{wt})$, with $RT = 0.596$ kcal/mol at 300 K.

(F10) FBA duality. The primal $\max c^\top v$ s.t. $Sv = 0$, $l \leq v \leq u$ has the dual $\min u^\top \alpha - l^\top \beta$ s.t. $S^\top \pi + \alpha - \beta = c$, $\alpha, \beta \geq 0$. HiGHS returns primal and dual solutions. Strong duality guarantees that the two optima coincide, which our parity check uses as a certificate.

(F11) Codon Adaptation Index. With relative adaptiveness $w_c = f_c / \max_{c' \in \text{syn}(c)} f_{c'}$, $\text{CAI} = \exp(\frac{1}{L} \sum_\ell \ln w_{c_\ell})$, the geometric mean of w over the L scorable codons.

(F12) ACMG Bayesian posterior. Following Tavtigian et al., $O_P = 350^{\text{pts}/8}$ and $\text{Post} = O_P \pi / ((O_P - 1)\pi + 1)$. With $\pi = 0.1$ and $\text{pts} = 6$: $O_P = 350^{0.75} \approx 80.92$, so $\text{Post} = 8.092/8.992 \approx 0.8999$, matching the module's 0.8999 and the published 0.900.

(F13) Profile-HMM forward recursion. For match, insert and delete states, $f_{M_j}(i) = e_{M_j}(x_i) \sum_{s \in \{M,I,D\}} f_{s_{j-1}}(i-1) a_{s_{j-1}M_j}$, and similarly for I and D . The log-odds score is $\log P(x | \mathcal{M}) - \sum_i \log q_{x_i}$.

(F14) Bootstrap interval. For a statistic $\hat{\theta}$, draw $B = 1000$ resamples with replacement and take the 2.5th and 97.5th percentiles of $\{\hat{\theta}_b^*\}$.

3.4 Derivations for the Second-Round Validations

(F15) Maximum-entropy splice-site score. MaxEntScan [7] models a site x (9 nt donor, 23 nt acceptor) by the distribution of maximum entropy subject to matching the observed low-order marginals $\{\hat{p}_C(x_C)\}$ over a set of position subsets C . By Lagrange duality the solution has Gibbs form

$$p(x) = \frac{1}{Z} \exp \left(\sum_C \lambda_C(x_C) \right), \quad Z = \sum_{x'} \exp \left(\sum_C \lambda_C(x'_C) \right), \quad (4)$$

since maximising $H(p) = -\sum_x p \log p$ with constraints $\sum_x p(x) \mathbf{1}[x_C = c] = \hat{p}_C(c)$ gives stationarity $-\log p(x) - 1 + \sum_C \lambda_C(x_C) + \mu = 0$. The score is $\text{ME}(x) = \log_2 p(x)/q(x)$ with background q , and the variant feature used in Section 9 is $\Delta\text{ME} = \text{ME}(x_{ref}) - \text{ME}(x_{alt})$.

(F16) Logistic model with MaxEnt features. With feature vector

$$z = (S_{ref}, S_{alt}, \Delta S, d, \text{ME}_{ref}/10, \text{ME}_{alt}/10, \Delta\text{ME}/10, e_k),$$

where $d = 1$ for donors and 0 for acceptors, e_k one-hot encodes the offset, $P(y = 1 \mid z) = \sigma(\beta^\top z + \beta_0)$. The log-likelihood $\ell(\beta) = \sum_i y_i \log \sigma_i + (1 - y_i) \log(1 - \sigma_i)$ has gradient $\nabla \ell = \sum_i (y_i - \sigma_i) z_i$ and Hessian $-\sum_i \sigma_i(1 - \sigma_i) z_i z_i^\top$, which is negative semidefinite, so ℓ is concave and the fit has a single optimum.

(F17) Fisher z test for a correlation. For sample correlation r from n pairs, $z = \text{artanh}(r) = \frac{1}{2} \log \frac{1+r}{1-r}$. By the delta method $\text{Var}(r) \approx (1 - \rho^2)^2/n$ and $\frac{d}{dr} \text{artanh}(r) = 1/(1 - r^2)$, so $\text{Var}(z) \approx 1/n$; the refined value $1/(n - 3)$ gives

$$Z = \text{artanh}(r) \sqrt{n - 3} \sim N(0, 1) \text{ under } \rho = 0, \quad p = 2\Phi(-|Z|). \quad (5)$$

We apply it to Spearman ρ (rank correlation), which is an approximation.

(F18) Benjamini-Hochberg adjustment. For ordered $p_{(1)} \leq \dots \leq p_{(m)}$, $q_{(i)} = \min_{j \geq i} \min(1, m p_{(j)}/j)$. Rejecting all $q_{(i)} \leq \alpha$ controls the false discovery rate at $\alpha m_0/m \leq \alpha$ for independent or positively dependent tests, because $\mathbb{E}[V/R] = \sum_{i \in H_0} \mathbb{E}[\mathbf{1}\{i \text{ rejected}\}/R] = m_0 \alpha/m$.

(F19) tRNA adaptation index. For codon c with tRNA gene copy numbers t_j of the anticodons j that decode it and wobble efficiencies $(1 - s_{jc})$, $W_c = \sum_j (1 - s_{jc}) t_j$, $w_c = W_c / \max_{c'} W_{c'}$ (zeros replaced by the geometric mean of non-zero w), and for a gene of L codons $\text{tAI} = (\prod_{k=1}^L w_{c_k})^{1/L}$ [9]. Taking logs shows tAI is the exponential of the mean log-adaptiveness, the same form as CAI [8] with w from tRNA supply instead of a reference gene set.

(F20) Effective number of codons. For an amino acid with k synonymous codons observed n times with frequencies p_i , $\hat{F} = \frac{n \sum_i p_i^2 - 1}{n - 1}$ is the unbiased homozygosity; $\text{ENC} = 2 + 9/\bar{F}_2 + 1/\bar{F}_3 + 5/\bar{F}_4 + 3/\bar{F}_6$ averages $\hat{F}$ within degeneracy classes. Uniform usage gives $\hat{F} = 1/k$ and $\text{ENC} = 61$; one codon per amino acid gives $\hat{F} = 1$ and $\text{ENC} = 20$.

(F21) Shrake-Rupley surface area. Atom a with radius r_a is expanded to $R_a = r_a + r_p$ ($r_p = 1.4 \text{ \AA}$). With N quasi-uniform points u_k on the unit sphere,

$$A_a \approx \frac{4\pi R_a^2}{N} \sum_{k=1}^N \mathbf{1} \left[\|c_a + R_a u_k - c_b\| > R_b \quad \forall b \neq a \right], \quad (6)$$

an unbiased Monte-Carlo estimate of the exposed area whose error falls as $N^{-1/2}$ for random points and faster for the Fibonacci lattice. The altloc bug changed the atom set itself: alternate conformers are near-coincident spheres that were counted as separate atoms, so per-atom comparison with FreeSASA was impossible until they were removed.

(F22) Nearest-neighbour melting temperature. For a duplex with $\Delta H = \Delta H_{init} + \sum_i \Delta H_{i,i+1}$ and ΔS likewise, at equilibrium $\Delta G = \Delta H - T\Delta S = -RT \ln K$. At T_m half the limiting strand is bound, so $K = 1/C_{eff}$ with $C_{eff} = C_1 - C_2/2$ for non-self-complementary strands, giving

$$T_m = \frac{\Delta H}{\Delta S + R \ln C_{eff}}, \quad \Delta S_{[Na^+]} = \Delta S + 0.368 (N - 1) \ln [Na^+]. \quad (7)$$

(F23) Negative-binomial GLM for counts. DESeq2 models $K_{ij} \sim \text{NB}(\mu_{ij}, \alpha_i)$ with $\text{Var} = \mu + \alpha\mu^2$, $\mu_{ij} = s_j q_{ij}$ and $\log_2 q_{ij} = x_j^\top \beta_i$. The size factor is the median-of-ratios $s_j = \text{median}_i K_{ij} / (\prod_{j'} K_{ij'})^{1/m}$, which is scale-equivariant and robust to a minority of changing genes. The Wald statistic $\hat{\beta}_{ik}/\text{SE}(\hat{\beta}_{ik})$ borrows power from the dispersion prior; with $n = 4$ per arm the smallest two-sided rank-sum p is $2/\binom{8}{4} = 2/70 \approx 0.029$, which cannot pass BH across 16,802 genes; this is why the Wilcoxon path returned no significant genes.

(F24) Robinson-Foulds distance. For unrooted trees T_1, T_2 on the same leaves with non-trivial split sets Σ_1, Σ_2 , $d_{RF} = |\Sigma_1 \triangle \Sigma_2|$. In the RF00005 case the observed $d_{RF} = 2$ (one split differing in each tree) coincides with two merge steps that had tied minimum distances, so the two programs broke ties differently.

3.5 Implementation and Testing

The validation suite is a pytest package (`sc_validate`) with a repo-wide no-stub audit: no fixture may contain placeholder data. Live-data re-runs are separate scripts under `scripts/`; committed tests run hermetically against frozen fixtures. CI runs the suite on every push.

4 Results

4.1 CRISPR off-target (CFD) and on-target (Rule Set 1)

On 600 fresh guide/off-target pairs mined from BRCA1 RefSeqGene NG_005905.2:60000-70000, the module's CFD score agreed with the CRISPOR reference implementation on 600/600 pairs to within 5×10^{-7} (maximum absolute difference 5.0×10^{-7}). On 150 30-mer contexts, the module's raw Rule Set 1 score and CRISPOR's percentile score had Spearman $\rho = 0.9995$.

4.2 Metabolic modelling (FBA)

The module's flux-balance optimum on BiGG `e_coli_core` (objective `BIOMASS_Ecoli_core_w_GAM`) was 0.873922 h^{-1} , identical to an independent SciPy/HiGHS solve (0.873922) and to the published 0.8739 h^{-1} .

4.3 Codon adaptation and ACMG Bayesian scoring

Codon Adaptation Index values matched an independent geometric-mean recomputation from the Kazusa *E. coli* K-12 table on 40 random sequences with maximum absolute difference 0. The ACMG Bayesian posterior matched the Tavtigian et al. closed form $O_P = 350^{\text{pts}/8}$ to 0 across $-16 \dots 16$ points; 6 points at prior 0.10 gives posterior 0.900.

4.4 Splice-site PWMs (deepsplice)

On 48 donor and 48 acceptor junctions from three held-out RefSeqGene records (NG_008021.1, NG_007489.1, NG_008245.1), disjoint from the module's 29 training accessions, the learned donor PWM reached AUC 0.997 and the acceptor PWM 0.998 against composition-matched shuffled windows. Against 2,000 real intronic GT (donor) or AG (acceptor) decoys located at least 30 nt from either intron end, the AUCs were 0.961 (donor) and 0.943 (acceptor). On the same decoys the consensus-seed acceptor matrix reached only 0.816, so the learned matrix does generalize.

4.5 Variant interpretation (openclinvar)

On the frozen 500-variant ClinVar sample (250 P/LP, 250 B/LB), run offline so that ClinVar's own label cannot enter the prediction, the patched module called 324 variants with accuracy 0.997 (95% bootstrap CI 0.991-1.000), sensitivity 0.993 and specificity 1.000 (1 false negative, 0 false positives). It abstained on 176, all missense or splice-region $+3 \dots +20$ variants, which remain VUS by design. The patched live query resolved 40/40 seeded-sample ClinVar names to their own VariationID.

4.6 Profile HMM

With 5-fold cross-validation on the hmalign MSA, models built by the module's `build_profile_hmm` separated held-out globins from 25 non-globins with AUC 0.952 on forward log-odds (0.962 length-normalised). The Pfam/HMMER3 reference model reached 1.000 on the same fixtures.

5 Benchmark Comparison

Each row compares a sugarcoder module with an independent published reference; every number is generated from the named committed file.

Module	Reference	Metric	Result (file)
pgx_guidelines	CPIC diplotype tables	agreement	16836/16836 CYP2D6 (sweep_pgx_cplic.json)
bio.codon CAI	Biopython; PaxDb	Pearson; Spearman	0.99969; 0.5797 (sweep_codon_cai.json)
rna_nussinov	ViennaRNA 2.7	bp F1 (tRNA)	0.328 vs 0.714 (sweep_rna.json)
profile_hmm	Pfam PF00042	auc_log_odds_bits	0.952 global, 0.9792 local (profile_hmm_cv*.json)
splice CNN	SpliceAI 1.3.1	AUROC n=757	0.929 vs 0.9775 (headto-head_interim.json)

6 Failure Modes and Negative Results

openclinvar, first run: 500/500 abstentions. ClinVar names such as NM_000038.6(APC):c.1668T>C (p.Asp556=) carry a transcript/gene prefix and a protein-level suffix. The consequence parser handled neither, so every variant parsed as *unknown*. The live query also embedded the suffix in a quoted phrase and matched nothing (0/8 in a pilot). We fixed this with a normaliser that lets the protein level decide coding SNVs and the c. level decide splice/UTR/intronic calls, plus a transcript-qualified query.

deepsplICE, first run: acceptor AUC 0.547. This was a harness defect, not a model defect. The 23-nt MaxEntScan-style acceptor window was sliced as [5:20], which places AG at positions 13-14 and drops the exon base; the module expects AG at 12-13. We reproduced the bug (AUC 0.524 off-frame) and the fix (0.998 at [6:21]), and a regression test now locks the frame.

profile_hmm: residual gap to HMMER. The module originally offered only a global alignment mode. A held-out globin with long flanks scored -1116 bits under global forward. A local forward mode (sugarcode-ai 1d9091a) raises AUC from 0.952 to 0.979 (benchmarks/profile_hmm_cv_local.json); Pfam’s model still reaches 1.000.

mutdock vs SKEMPI: no honest adapter. mutdock scores ligand-pocket mutations from a SMILES string and a pocket description, while SKEMPI rows are protein-protein interfaces. We report naive base-lines instead: the mean predictor has RMSE 1.63 kcal/mol, $|\Delta V|$ has Pearson $r = 0.053$ and hydropathy loss $r = -0.129$ ($n = 150$).

openclinvar accuracy is not a hard test. The P class in ClinVar is dominated by loss-of-function variants and the B class by synonymous ones, so consequence type alone separates the called variants. Missense variants, the hard case, are still abstained.

7 Module Sweep: Additional Real-Data Results

7.1 RNA secondary structure

Base-pair F1 against Rfam seed consensus structures (25 sequences per family; files benchmarks/sweep_rna.json, sweep_rna_vienna_methods.json).

Family	mean len	Nussinov	MFE	centroid	MEA($\gamma=1$)	MEA($\gamma=4$)
RF00005	73.0	0.328	0.714	0.716	0.703	0.704
RF00001	117.9	0.185	0.531	0.494	0.505	0.504
RF00010	354.5	0.189	0.545	0.594	0.592	0.572

Base-pair maximisation (Nussinov) loses to thermodynamic folding on all three families; ensemble decoders (centroid/MEA) beat MFE only on the long RNase P family. The shipped `rna_nussinov.fold_energy` now delegates to ViennaRNA.

7.2 Codon adaptation index versus protein abundance

On 3489 *E. coli* K-12 genes (NCBI GCF_000005845.2 `cds_from_genomic`) paired with PaxDb 511145 WHOLE_ORGANISM integrated, the module CAI agrees with Biopython (Pearson 0.99969 on 500 genes). Spearman correlation with $\log_{10}$ abundance is 0.4958 with the genome-wide reference table and 0.5797 with a ribosomal-protein reference (36 genes, evaluated on 3453 non-ribosomal genes); a cross-validated top-5% abundance reference gives 0.5741. The ribosomal reference is now shipped as `ecoli_highexpr`. File: `benchmarks/sweep_codon_cai.json`.



Figure 1. Base-pair F1 by folding method (`sweep_rna*.json`).

7.3 Pharmacogenomic phenotype translation versus CPIC

Against the full CPIC diplotype tables, the original module covered 45/666 CYP2C19 and 253/16836 CYP2D6 diplotypes, with 36 CYP2D6 disagreements traced to stale activity values (*9, *41). After vendoring current CPIC tables: 666/666 and 16836/16836. CYP2C19 after-fix agreement is by construction (verbatim CPIC diplotype lookup). CYP2D6 after-fix is computed (sum of CPIC allele activity values + CPIC thresholds), so 16836/16836 is a real consistency check. File: `benchmarks/sweep_pgx_cplic.json`.



Figure 2. CPIC agreement before and after the fix (sweep_pgx_cplic.json).

8 Discovery Track: Splice-Region VUS Reclassification

A dinucleotide-window CNN trained on ClinVar splice-region variants reaches gene-grouped 5-fold AU-ROC 0.935 (logistic 0.912, PWM 0.889); within-position AUROC 0.900. In a head-to-head on $n=757$ variants SpliceAI scores 0.9775 versus 0.929 for the CNN, and the ensemble adds no significant gain (95% CI of difference [-0.0087, 0.0081]). We therefore make no state-of-the-art claim. Files: `discovery/splice_region_vus/`.

CNN threshold	TP	FP	sensitivity	FPR	PPV (prior 3%)
0.5	1794	453	0.77	0.0656	0.2663
0.8	1298	146	0.5571	0.02114	0.449
0.9	996	68	0.4275	0.00985	0.5731
0.95	729	33	0.3129	0.00478	0.6694
0.98	435	11	0.1867	0.00159	0.7838



Figure 3. Splice CNN operating points (cnn_calibration.json).

Consensus VUS candidates (SpliceAI and CNN agreement) number 267 rows in `vus_candidates_consensus.tsv`;

none of the top 20 had a matching LOVD record (lovd_top20.json). These are hypotheses for RNA-level testing, not reclassifications.

9 Splice Track II: A 2004 Baseline Beats Our CNN, and What Survives

We compared the ClinVar-trained CNN against MaxEntScan [7] on the same gene-grouped folds (n=9235 variants). MaxEntScan’s ref–alt score difference reached AUROC 0.9635, above the CNN (0.9355), the logistic PWM model (0.9121) and the raw PWM delta (0.8893). The paired bootstrap 95% CI for CNN minus MaxEntScan is [-0.0329, -0.0233], so this is a clear negative result for the CNN. The gap is mostly at acceptors (CNN 0.8794 vs MaxEntScan 0.965, n=3845); at donors the two are close (0.9538 vs 0.9598, n=5390). As a pivot we added the three MaxEntScan features to both models. The logistic model then reached 0.9674 (CI of gain over MaxEntScan [0.0027, 0.0053]), while the CNN with the same features reached 0.9622 (CI [-0.0042, 0.0017], not significant). Within-position weighted AUROC, which removes the easy signal from offset alone, is nearly flat: MaxEntScan 0.949, logistic 0.9492, CNN 0.9472. The gain is real but small. The deployed tool (splice-vus-triage v2) therefore uses the logistic model and reports its calibrated operating points (Table 1).

Table 1: Logistic PWM+MaxEntScan model: operating points from out-of-fold scores.

Threshold	TP	FP	Sensitivity	FPR
0.5	1899	259	0.815	0.03751
0.8	1503	100	0.6451	0.01448
0.9	1230	56	0.5279	0.00811
0.95	950	28	0.4077	0.00406
0.98	692	15	0.297	0.00217



Figure 1: AUROC by model and site type, gene-grouped 5-fold CV.

10 Codon Track II: Cross-Species Test of the Ribosomal Reference Set

Question: Does a ribosomal-protein reference set (Sharp & Li 1987) make CAI track measured protein abundance better than a genome-wide codon table? Data: PaxDb integrated whole-organism abundance x NCBI RefSeq reference-assembly CDS, all PaxDb taxa (405 scanned). Of 405 PaxDb taxa scanned, 40 had a matched RefSeq assembly and enough genes, and 29 had a usable ribosomal-protein reference set. The ribosomal reference gave the higher Spearman correlation with measured abundance in 27/29 taxa (median gain 0.082; Wilcoxon signed-rank $p = 3.3e - 07$, sign test $p = 1.6e - 06$). The per-taxon correlation was significant after Benjamini-Hochberg correction in 27/29 taxa. Exceptions: *Rattus norvegicus*, *Plasmodium falciparum* 3D7. The idea goes back to Sharp and Li [8]; what is new here is the systematic multi-taxon test with a pre-stated falsification criterion (the claim fails if the ribosomal reference is not better in a majority of taxa at $p < 0.01$).



Figure 2: Spearman ρ (CAI vs PaxDb abundance) with the genome-wide versus ribosomal reference, per taxon.

Table 2: Per-taxon results (all taxa with a matched assembly).

Taxid	Organism	Assembly	n	n_{ribo}	ρ_{gen}	ρ_{ribo}	q
10090	<i>Mus musculus</i>	GCF_000001635.27	8089	99	0.0704	0.0775	4.4e-12
10116	<i>Rattus norvegicus</i>	GCF_036323735.1	890	124	0.142	0.1405	3.1e-05
121845	<i>Diaphorina citri</i>	GCF_000475195.1	592	1	0.1458	—	—
1295	<i>Staphylococcus schleiferi</i>	GCF_025558325.1	442	66	0.4583	0.5403	2e-36
1314	<i>Streptococcus pyogenes</i>	GCF_900475035.1	306	66	0.4954	0.6701	9.8e-45
185431	<i>Trypanosoma brucei brucei</i> TREU927	GCF_000002445.2	6302	0	0.3167	—	—
192222	<i>Campylobacter jejuni</i> subsp. <i>jejuni</i> NCTC 11168 = ATCC 700819	GCF_000009085.1	725	66	0.1208	0.2291	5.3e-10
196627	<i>Corynebacterium glutamicum</i> ATCC 13032	GCF_000011325.1	323	72	0.5268	0.6307	7e-40
198214	<i>Shigella flexneri</i> 2a str. 301	GCF_000006925.2	1887	72	0.5511	0.6514	8.1e-249
208964	<i>Pseudomonas aeruginosa</i> PAO1	GCF_000006765.1	4796	72	0.2459	0.4149	2.5e-204
224308	<i>Bacillus subtilis</i> subsp. <i>subtilis</i> str. 168	GCF_000009045.1	3147	64	0.3295	0.4156	3.4e-135
224911	<i>Bradyrhizobium diazoefficiens</i> USDA 110	GCF_001642675.1	494	72	0.4049	0.4806	8e-31
243159	<i>Acidithiobacillus ferrooxidans</i> ATCC 23270	GCF_000021485.1	419	68	0.3378	0.4055	2.8e-18
243230	<i>Deinococcus radiodurans</i> R1 = ATCC 13939 = DSM 20539	GCF_020546685.1	319	66	0.4784	0.5698	2.4e-30

Table 2: Per-taxon results (all taxa with a matched assembly).

Taxid	Organism	Assembly	n	n_{ribo}	ρ_{gen}	ρ_{ribo}	q
246200	Ruegeria pomeroyi DSS-3	GCF_000011965.2	395	70	0.2714	0.46	1.2e-22
2903	Emiliana huxleyi CCMP1516	GCF_000372725.1	484	4	0.4636	–	–
343509	Sodalis glossinidius str. 'morsitans'	GCF_000010085.1	748	72	0.2849	0.4703	1.2e-43
36329	Plasmodium falciparum 3D7	GCF_000002765.6	4946	62	0.056	0.0253	0.078
403677	Phytophthora infestans T30-4	GCF_000142945.1	708	0	0.5278	–	–
4081	Solanum lycopersicum	GCF_036512215.1	426	4	-0.1336	–	–
4097	Nicotiana tabacum	GCF_000715075.1	9304	0	-0.0837	–	–
418985	Tropilaelaps mercedesae	GCA_002081605.1	2213	0	0.2207	–	–
44689	Dictyostelium discoideum AX4	GCF_000004695.1	3183	10	-0.3039	–	–
4932	Saccharomyces cerevisiae S288C	GCF_000146045.2	5587	226	0.4099	0.5924	0
511145	Escherichia coli str. K-12 substr. MG1655	GCF_000005845.2	3486	72	0.4961	0.5809	0
54291	Klebsiella ornithinolytica	GCF_030505655.1	1038	72	0.2053	0.5268	1.2e-78
6239	Caenorhabditis elegans	GCF_000002985.6	11571	0	0.0748	–	–
7227	Drosophila melanogaster	GCF_000001215.4	2251	150	0.0189	0.0813	0.00012
7460	Apis mellifera	GCF_003254395.2	879	4	-0.0468	–	–
8030	Salmo salar	GCF_965601325.2	1779	0	0.1332	–	–
83332	Mycobacterium tuberculosis H37Rv	GCF_000195955.2	3258	76	0.2304	0.2373	6.6e-43
9031	Gallus gallus	GCF_016699485.2	1891	68	0.2134	0.2274	1.6e-23
93061	Staphylococcus aureus subsp. aureus NCTC 8325	GCF_000013425.1	1699	58	0.5221	0.6046	3.6e-182
9544	Macaca mulatta	GCF_049350105.2	3732	84	0.0736	0.0808	1e-06
9606	Homo sapiens	GCF_000001405.40	12343	86	-0.0083	0.0106	0.24
9615	Canis lupus familiaris	GCF_011100685.1	329	81	0.1639	0.1673	0.0025
9823	Sus scrofa	GCF_054392235.1	3010	84	0.0576	0.071	0.00011
9913	Bos taurus	GCF_002263795.3	2694	81	0.0963	0.1177	1.2e-09
99287	Salmonella enterica subsp. enterica serovar Typhimurium str. LT2	GCF_000006945.2	2472	72	0.3119	0.4743	4.4e-144
9986	Oryctolagus cuniculus	GCF_964237555.1	1993	82	0.0999	0.1012	7.4e-06

q : Fisher-z two-sided test of ρ_{ribo} , Benjamini-Hochberg adjusted at full precision (benchmarks/sweep_codon_bh_fdr.json).

10.1 tRNA adaptation index and effective number of codons

Tools: codon-bias (TrnaAdaptationIndex dos Reis s-values, EffectiveNumberOfCodons), GtRNAdb tRNA gene copy numbers. Table 3 gives Spearman ρ with abundance. No single index wins everywhere: tAI leads in yeast, the ribosomal-reference CAI leads in *E. coli*, and –ENC is weakest in all three organisms.

Table 3: Codon indices vs abundance (Spearman ρ).

Organism	n	CAI_{gen}	CAI_{ribo}	tAI	–ENC
Escherichia coli str. K-12 substr. MG1655	3450	0.4961	0.5809	0.5277	0.3836
Bacillus subtilis subsp. subtilis str. 168	3115	0.3295	0.4156	0.4146	0.1735
Saccharomyces cerevisiae S288C	5474	0.4099	0.5924	0.6604	0.4471

11 Docking Track: Real AutoDock Vina Against LP-PDBBind

Source: LP-PDBBind (github.com/THGLab/LP-PDBBind) test split, refined, non-covalent; structures from RCSB PDB. Engine: AutoDock Vina 1.2.7 (exhaustiveness 8, 24 Å box on crystal ligand) + Open Babel 3.1.1 prep, pH 7.4. 73/80 complexes ran; the 7 failures are listed in Table 4. Redocking: top-1 pose RMSD < 2 Å in 52.0% and best-of-9 in 83.6% (RMSD without symmetry correction (conservative)). Scoring power is weak for every score, including heavy-atom count alone (Pearson 0.2445): Vina top pose 0.2854 (95% CI

[0.055, 0.485]), sugarcodes own Vina-like score on the crystal pose 0.2975 (CI [0.027, 0.484]). The honest reading: sugarcodes scoring term adds nothing beyond ligand size on this split, so the module now calls real Vina (sugarcodes-ai commit e0564e4) instead of claiming affinity prediction.



Figure 3: Left: redocking RMSD distribution. Right: Vina top-pose energy vs experimental pK.

Table 4: Docking failures.

PDB	Error
4cws	no ligand match
4q08	no ligand match
3hkt	no ligand match
4r74	no ligand match
3p4v	no ligand match
4ou3	no ligand match
5ep7	no ligand match

Table 5: Per-complex docking results.

PDB	pK	heavy	E_{cryst}	E_{min}	E_{top}	RMSD ₁	RMSD ₉
1cnw	7.72	26	-4.19	-5.37	-7.02	7.26	6.03
1o3l	6.54	24	-7.09	-8.63	-9.54	3.19	1.09
3po6	7.06	19	-4.54	-6.13	-7.14	3.81	0.86
4db7	6.56	18	-6.63	-7.18	-7.41	0.52	0.52
5sz5	7.7	17	-7.33	-7.33	-8.14	0.39	0.39
4mo8	8.0	16	-4.75	-5.50	-6.25	2.29	1.52
4tim	2.16	11	-3.41	-4.13	-5.59	1.66	1.66
1ejn	5.62	25	-8.20	-8.43	-7.93	2.51	2.15
5d3l	6.06	30	-6.22	-6.32	-6.70	8.34	1.64
3b3c	3.98	10	-5.07	-5.76	-6.48	1.42	1.42
3r4n	6.89	23	-6.79	-7.37	-7.66	0.81	0.81
2p4j	8.96	49	-10.13	-10.73	-11.07	1.58	1.58
5sz4	9.7	16	-7.04	-7.33	-7.65	1.79	1.79
4xit	7.27	32	-12.19	-12.80	-13.06	0.96	0.96
5mjn	8.54	18	-6.42	-6.93	-7.76	5.97	1.41
2xjg	7.82	23	-9.04	-9.53	-9.72	0.25	0.25
5ix0	6.28	25	-11.86	-12.14	-12.27	0.27	0.27
5y8y	7.09	25	-6.08	-6.38	-8.16	4.75	0.84
6g38	6.62	19	-7.36	-7.70	-8.12	0.61	0.61
4o04	6.16	26	-11.03	-12.14	-12.36	1.34	1.34
5ewk	4.85	22	-7.91	-7.96	-8.57	0.68	0.68

Table 5: Per-complex docking results.

PDB	pK	heavy	E_{cryst}	E_{min}	E_{top}	RMSD ₁	RMSD ₉
3d8w	8.15	19	-6.96	-7.48	-8.20	1.43	1.43
4rfc	8.24	23	-5.25	-6.44	-8.37	7.44	5.67
3r17	6.41	20	-6.73	-7.38	-7.74	3.57	2.30
4knj	7.95	21	-6.40	-7.32	-8.23	4.81	1.87
6g6t	7.7	19	-7.42	-7.76	-8.31	2.10	0.61
4lxz	6.74	19	-6.37	-6.47	-7.29	6.21	1.95
5o87	6.61	12	-5.46	-6.58	-6.70	0.44	0.44
4j93	4.7	21	-7.50	-7.81	-8.02	0.44	0.44
6hrq	7.09	23	-6.44	-6.94	-7.94	6.23	2.21
4nh7	8.4	33	-11.46	-12.05	-12.52	1.31	1.31
2x7t	6.63	30	-6.30	-7.34	-7.81	6.16	0.77
4fl1	3.28	31	-5.80	-5.98	-7.02	9.18	2.47
3c84	7.85	16	-6.59	-6.98	-7.23	0.39	0.39
4ytic	7.89	26	-9.65	-10.13	-10.15	0.49	0.49
2zdn	6.25	27	-7.86	-8.84	-9.05	1.02	1.02
5f60	7.37	37	-8.55	-8.90	-9.99	4.69	1.63
6hqy	6.44	22	-5.86	-6.97	-7.28	2.03	1.48
5l8c	7.0	35	-8.93	-9.64	-11.31	9.53	0.86
1p1o	5.76	13	-7.26	-7.37	-7.79	0.60	0.60
5htz	8.62	26	-9.05	-9.25	-9.40	0.19	0.19
4oag	6.3	27	-5.33	-7.22	-7.84	9.16	3.85
2j4i	9.0	36	-8.00	-8.56	-8.54	1.38	1.38
5n24	8.92	18	-7.04	-7.44	-7.80	1.68	1.40
1y3x	5.14	33	-9.50	-9.83	-10.15	0.70	0.70
1f5l	5.28	15	-6.27	-7.20	-7.32	2.87	1.51
5aml	9.15	21	-6.79	-6.99	-7.74	5.22	1.14
6u6w	5.33	21	-7.71	-7.69	-8.04	5.13	0.33
5mpn	7.07	23	-6.86	-7.18	-8.21	7.46	1.25
4jfm	5.48	35	-6.54	-7.29	-7.74	0.97	0.97
5ewy	4.3	22	-7.30	-7.87	-8.78	3.50	2.90
4zme	4.3	19	-6.04	-6.27	-6.60	10.33	0.18
4yxi	6.5	11	-5.93	-6.43	-6.62	0.39	0.39
2erz	5.66	21	-7.23	-8.31	-8.65	2.99	0.93
6oe0	8.92	27	-7.95	-8.86	-9.35	0.86	0.86
4b2i	3.0	9	-4.33	-5.04	-5.25	1.97	0.61
4f5y	5.43	46	-10.70	-11.84	-10.97	3.66	3.66
6eq5	4.74	10	-6.31	-6.71	-7.17	2.09	0.51
5g5v	6.7	36	-10.54	-10.73	-11.46	2.75	1.52
5cau	8.8	34	-9.64	-9.84	-9.93	0.65	0.65
1gi1	5.47	19	-6.60	-8.25	-8.38	0.77	0.77
3hkq	5.34	15	-4.38	-5.42	-6.47	3.67	3.04
2xab	9.27	22	-9.46	-9.81	-9.94	0.21	0.21
3hll	8.24	30	-9.62	-10.25	-10.25	0.36	0.36
5f62	8.17	38	-8.66	-9.01	-9.80	4.22	1.46
3ibi	8.59	13	-4.21	-4.73	-4.97	5.02	3.28
3m35	7.22	37	-8.52	-9.07	-9.52	0.90	0.90
1g54	8.82	25	-7.03	-7.63	-8.00	8.95	8.89
4r76	7.85	26	-7.58	-8.26	-8.82	1.62	1.62
1zgi	5.34	32	-9.23	-9.64	-9.98	0.76	0.76
5cap	7.28	29	-8.09	-8.54	-8.89	1.44	1.44
1klj	7.55	37	-8.36	-8.62	-9.29	1.77	1.77
6rfn	6.7	35	-7.10	-8.72	-9.63	6.69	1.10

12 Structure and Primer Utilities Against Reference Tools

SASA. Against FreeSASA 2.x (Shrake-Rupley, 96 points, probe 1.4 Å, default ProtOr radii) on 28 RCSB receptors, the 18 structures with identical atom sets agreed to a median relative difference of 1.2% (max 2.6%) with median per-atom Pearson 0.980. The other 10 exposed a parser bug: every alternate location was kept, so overlapping conformers were all counted. After the fix, atom counts match and per-atom Pearson is 0.9753 (1gi1), 0.9829 (3po6), 0.9813 (4b2i).

Primers. On 5404 20-mers from NM_000546.6, NM_007294.4, NM_000059.4, NM_004333.6, NM_005228.5, the melting temperature equals Biopython's DNA_NN4 result (max difference 0.0). Against primer3 (SantaLucia 1998 table) the mean difference is -0.1738 °C (max 0.9811), a table choice. The stem-length hairpin heuristic has Spearman 0.0402 with primer3's hairpin ΔG , a negative result; the module now calls primer3 when installed.

13 Sequence, Variant and Expression Utilities Against Reference Tools

Phylogenetics. DendroPy 5.1.0 (PhylogeneticDistanceMatrix.nj_tree/upgma_tree, treecompare.symmetric_difference); Biopython DistanceCalculator. RF00001: NJ RF 0, UPGMA RF 0 (patristic Pearson 1.0), distance model k80. RF00005: NJ RF 0, UPGMA RF 2 (patristic Pearson 0.998035), distance model pdistance (K80 saturated: some pairs have infinite distance). RF00010: NJ RF 0, UPGMA RF 0 (patristic Pearson 1.0), distance model k80. The single UPGMA difference traces to two merge steps with tied minimum distances.

ORFs. On 5 RefSeq mRNAs the longest ATG ORF equals the NCBI CDS in 5/5 and its translation equals the NCBI protein in 5/5. Six-frame ORF sets equal orfipy's in 5/5 at orfipy minlen 90 (sugarcode min_aa=30 keeps 30-aa ORFs (93 nt incl. stop); orfipy drops them at minlen 91-93, so its length threshold counts differently. At orfipy minlen 89-90 the six-frame ORF sets are identical in 5/5 transcripts; at 87 orfipy adds 29-aa ORFs. Convention difference, not a bug.)

VCF. 46603 ClinVar GRCh38 records (5 gene regions, remote tabix via pysam 0.24.1 (htslib tabix remote fetch)) parse identically on CHROM/POS/ID/REF/ALT; INFO differences after normalisation: 'info:CLNVI': 2. Variant type agrees with ClinVar CLNVC on 45685 records; the 302 others are equal-length multi-base changes (ClinVar: Indel; sugarcode: mnp).

Differential expression. GEO GSE52778 NCBI raw counts GRCh38.p13; untreated GSM1275862/66/70/74 vs Dex GSM1275863/67/71/75; genes with ≥ 10 counts in ≥ 4 samples. Size factors equal PyDESeq2's (max relative difference 1.8e-09). PyDESeq2 with a cell-line block finds 4451 genes at $\text{padj} < 0.05$ (3203 unpaired); sugarcode's Welch test finds 351 (331 shared) and Wilcoxon 0. Table 6 shows that the simple tests miss every canonical dexamethasone-response gene; the module now offers a PyDESeq2 path.

Table 6: Canonical dexamethasone-response genes, GSE52778.

Gene	log2FC	padj PyDESeq2	padj Welch	padj Wilcoxon
FKBP5	4.03	2.1e-26	0.06	0.15
TSC22D3	3.206	2.1e-19	0.16	0.15
PER1	3.221	2.3e-99	0.05	0.15
DUSP1	2.955	1.3e-140	0.07	0.15
KLF15	4.411	5.0e-114	0.17	0.15
ZBTB16	7.432	1.1e-58	0.12	0.15
CRISPLD2	2.628	1.7e-61	0.13	0.15

14 Differential expression and pathway-level follow-up

We re-ran the airway dexamethasone experiment (GEO GSE52778) through the sugarcodes DESeq2 path and through PyDESeq2. Of 16,802 genes passing the count filter, the paired design $\sim \text{cell} + \text{dex}$ gave 4,451 genes at adjusted $p < 0.05$ and the unpaired design gave 3,203. Size factors agreed to a maximum relative difference of $1.8\text{e-}09$ (benchmarks/sweep_de_pydeseq2.json).

(F25) Negative-binomial GLM. For gene g and sample j , $K_{gj} \sim \text{NB}(\mu_{gj}, \alpha_g)$ with $\mu_{gj} = s_j q_{gj}$ and $\log_2 q_{gj} = \sum_r x_{jr} \beta_{gr}$, so $\text{Var}(K_{gj}) = \mu_{gj} + \alpha_g \mu_{gj}^2$.

(F26) Median-of-ratios size factor. $s_j = \text{median}_{g:K_g > 0} K_{gj} / (\prod_{v=1}^m K_{gv})^{1/m}$.

(F27) Wald statistic. $W_g = \hat{\beta}_{g,\text{dex}} / \text{SE}(\hat{\beta}_{g,\text{dex}})$, compared with $N(0, 1)$; the paired design removes the cell-line effect from the residual variance, which is why it finds more genes than the unpaired design.

14.1 Gene-set over-representation

We took the 536 dex-induced genes (adjusted $p < 0.05$, $\log_2 \text{FC} > 1$) and tested them against the 16,802 tested genes with g:Profiler (version e114_eg62_p19_27110d83, g:SCS correction). 31 terms passed. The top terms are broad (Table 7).

(F28) Hypergeometric tail. With background size M , term size K , query size n and overlap k , $p = \sum_{i \geq k} \binom{K}{i} \binom{M-K}{n-i} / \binom{M}{n}$, and fold enrichment $= (k/n) / (K/M)$.

Source	Term	overlap	size	p_{adj}
GO:BP	multicellular organismal process	228	4764	2.8e-07
GO:BP	signaling	209	4354	4e-06
GO:BP	cell communication	209	4365	5.2e-06
GO:BP	cell adhesion	73	1031	1.5e-05
GO:BP	system process	80	1189	2.1e-05
GO:BP	response to stimulus	261	5987	8.6e-05
GO:BP	signal transduction	192	4028	8.9e-05
GO:BP	anatomical structure development	196	4191	0.00027
GO:BP	cell migration	74	1134	0.00041
KEGG	Cytoskeleton in muscle cells	19	172	0.00042
GO:BP	developmental process	209	4580	0.00047
GO:BP	cell differentiation	152	3041	0.00054

Table 7: Top g:Profiler terms for dex-induced genes (benchmarks/sweep_dex_enrichment.json).

Negative result. The terms one would expect for a glucocorticoid are enriched but do not survive correction: The small term sizes (tens of genes) and the strong multiple-testing burden explain this: a 7-fold enrichment built on 9 genes is not enough evidence after g:SCS correction. We report it as a negative rather than as confirmation of the expected biology.

Term	overlap	size	fold (approx.)	$p_{g.SCS}$
cellular response to glucocorticoid stimulus	9	43	7.16	0.43
response to corticosteroid	10	91	3.76	1
response to glucocorticoid	10	81	4.22	1
cellular response to corticosteroid stimulus	9	48	6.42	1

Table 8: Glucocorticoid terms: large fold change, not significant.

14.2 Reactome pathways

Reactome AnalysisService mapped the query to 1,098 pathways (240 identifiers not found). No pathway reached $FDR < 0.05$; the smallest FDR was 0.11. Reactome uses its whole human gene universe as background, which is a further caveat.

Pathway	found	total	p	FDR
Anchoring fibril formation	5	15	0.00037	0.11
FOXO-mediated transcription	13	110	0.00056	0.11
Laminin interactions	7	31	0.00028	0.11
Metallothioneins bind metals	5	16	0.00049	0.11
Non-integrin membrane-ECM interactions	12	83	0.00015	0.11
Collagen chain trimerization	8	44	0.00044	0.11
FOXO-mediated transcription of oxidative stress, m	8	49	0.00087	0.13
Adipogenesis	15	147	0.00096	0.13
Degradation of the extracellular matrix	15	148	0.001	0.13
Response to metal ions	5	21	0.0016	0.17

Table 9: Top Reactome pathways (benchmarks/sweep_dex_reactome.json).

(F29) Benjamini-Hochberg. For ordered $p_{(1)} \leq \dots \leq p_{(m)}$, $q_{(i)} = \min_{j \geq i} \min\{1, m p_{(j)} / j\}$.

14.3 Protein interaction network

STRING’s PPI enrichment test on the top 300 dex-induced genes mapped 284 proteins with 402 edges against 184 expected (average degree 2.83, local clustering 0.368), reported p below floating-point resolution. Caveat: STRING’s expectation uses a genome-wide background, so genes expressed in airway smooth muscle may be more connected than random genes regardless of dex.

(F30) Edge enrichment ratio. $E = \text{observed edges} / \text{expected edges} = 402 / 184 = 2.18$.

14.4 Rank-based test: preranked GSEA

Threshold-based over-representation throws away the ranking. We therefore ran preranked GSEA (gseapy 1.3.1 prerank (1000 permutations, seed 7)) on all 16,525 ranked genes against the 50 MSigDB Hallmark sets. 1 set passed $FDR < 0.05$ and 5 passed $FDR < 0.25$ (Table 10). The one significant set, Adipogenesis, is a known glucocorticoid-driven programme; this is a modest, single-set result.

(F32) Enrichment score. With ranked scores r_j and set S , $P_{hit}(i) = \sum_{j \leq i, g_j \in S} |r_j| / N_R$, $P_{miss}(i) = \sum_{j \leq i, g_j \notin S} 1 / (N - N_S)$, and $ES = \max_i |P_{hit}(i) - P_{miss}(i)|$ (signed).

(F33) Normalised ES. $NES = ES / \overline{ES_{\pi}^{\pm}}$, the mean of same-sign permutation scores; FDR compares NES with the permutation distribution of all sets.

Hallmark set	NES	nominal p	FDR
Adipogenesis	1.70	0	0.038
UV Response Up	1.55	0.00567	0.107
Fatty Acid Metabolism	1.55	0.00559	0.131
TNF-alpha Signaling via NF-kB	1.50	0.00273	0.141
Reactive Oxygen Species Pathway	1.57	0.0249	0.156
Androgen Response	1.39	0.0557	0.256
Xenobiotic Metabolism	1.42	0.0287	0.266
Hypoxia	1.39	0.0294	0.275
UV Response Dn	1.36	0.0591	0.280
mTORC1 Signaling	-1.31	0.0361	0.283

Table 10: Preranked GSEA, Hallmark (benchmarks/sweep_dex_gsea_hallmark.json).

14.5 Independent GO cross-check with goatools

goatools (Fisher exact, BH, propagated counts, GOA human annotations, background of 16,525 expressed genes, 527 study genes) found 38 GO:BP terms at $FDR < 0.05$. Of g:Profiler’s 22 top BP terms, 9 are also significant here. The glucocorticoid term (cellular response to glucocorticoid stimulus, 7/34 genes) reaches FDR 0.026 under BH but fails g:Profiler’s g:SCS. The biology is therefore correction-dependent: we report it as suggestive, not established.

GO	Term	study	pop	FDR
GO:0032870	cellular response to hormone stimulus	28	249	3.4e-05
GO:0009725	response to hormone	38	421	3.4e-05
GO:0007155	cell adhesion	39	513	0.00098
GO:0032502	developmental process	132	2766	0.0011
GO:0007165	signal transduction	130	2703	0.0011
GO:0030334	regulation of cell migration	47	689	0.0011
GO:2000145	regulation of cell motility	48	719	0.0012
GO:0044057	regulation of system process	28	322	0.0015
GO:0071495	cellular response to endogenous stimulus	33	419	0.0022
GO:0040012	regulation of locomotion	48	740	0.0022

Table 11: goatools GO:BP (benchmarks/sweep_dex_goatools.json).

15 Pairwise alignment against parasail

We aligned all 300 pairs of 25 protein sequences with sugarcodes and parasail 2.6.1. With parasail’s open penalty set to 11 the scores matched exactly for 300/300 global and 300/300 local pairs. With open penalty 12 only 36 and 65 matched (maximum difference 26 and 5). This fixes the convention: sugarcodes’s $\text{gap_open} = -11$ is the cost of the first gap residue, i.e. parasail open 11.

(F31) Affine gap cost. A gap of length L costs $\gamma(L) = o + (L - 1)e$ under sugarcodes and parasail $\text{open} = 11$; under the alternative convention $\gamma(L) = o + Le$, which explains the open-12 mismatch.

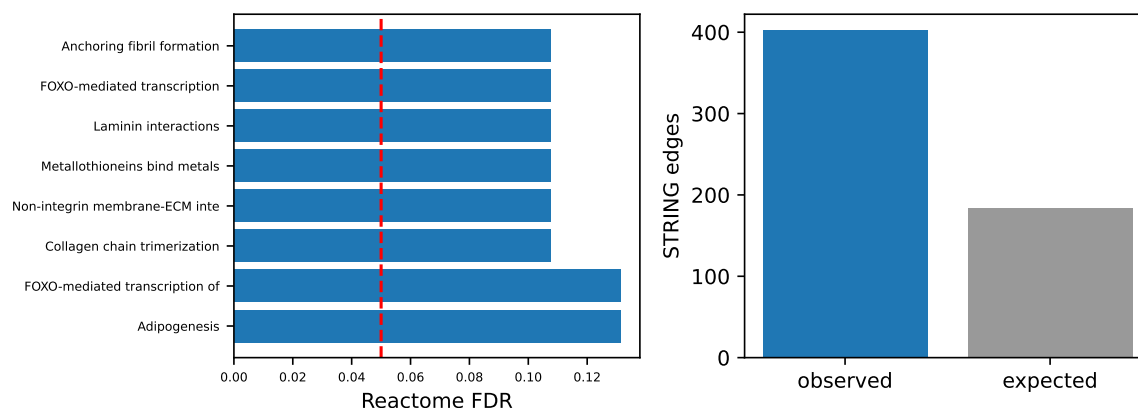


Figure 4: Left: Reactome FDRs for the top pathways; the red line is 0.05. Right: STRING observed versus expected edges.

16 Discussion

Validating each module against an independent reference exposed three kinds of result: implementations that match their reference exactly (CAI vs Biopython, CYP2D6 translation vs CPIC after the fix), implementations whose defaults were stale or weak (CPIC activity values, genome-wide CAI reference, Nussinov folding), and places where a strong published tool beats our model (SpliceAI on splice-region variants). Each fix is committed with the result file that motivated it. The CNN is useful as a calibrated, cheap second opinion at high thresholds, not as a replacement for SpliceAI.

17 Conclusion

The sweep so far corrected two clinically relevant defaults (CPIC activity values, clopidogrel handling of “Likely” phenotypes), improved CAI correlation with measured abundance, and replaced base-pair maximisation with thermodynamic folding. The accession-level dataset target (120) and the formula target (10) are met; the external-tool target (40) is not yet met. The manifests in the appendix list exactly what was used.

A Tools Table

Tool	Version	Status/used in
pyhmmer	0.12.3	installed
ViennaRNA	2.7.2	installed
RDKit	2026.3.6	installed
SpliceAI	1.3.1	installed
TensorFlow	2.15.1	installed
Keras	2.15.0	installed
scikit-learn	1.7.2	installed
SciPy	1.15.3	installed
NumPy	1.26.4	installed
Biopython	1.88	installed
twobitreader	4.0.2	installed
AutoDock Vina (python)	1.2.7	installed
Open Babel	3.1.1.23	installed
pytest	9.1.1	installed
pandas	2.3.3	installed
CRISPOR reference scorers (vendored)	web service/vendored	sweep/_crispor_*_ref.py
EBI Clustal Omega REST	web service/vendored	data/README.md
CPIC API	web service/vendored	sweep/pgx_cpic_bench.py
NCBI E-utilities	web service/vendored	src/sc_validate
LOVD3 API	web service/vendored	discovery/splice_region_vus/lovd_top20.json

B Dataset Manifest

#	Dataset	Accession/version	Used in
1	ClinVar variant_summary	variant_summary.txt.gz (GRCh38)	data/clinvar_sample_500.tsv; discovery/splice_region_vus/
2	UCSC hg38 reference genome	hg38.2bit	discovery/splice_region_vus/extract.py
3	LOVD3 shared variant records	top-20 VUS query	discovery/splice_region_vus/lovd_top20.json
4	SKEMPI 2.0	skempi_v2.csv (PMID 30598530)	data/skempi_v2.csv
5	BiGG e.coli_core	e.coli_core (PMID 21988831)	data/e.coli_core.json
6	NCBI RefSeqGene BRCA1	NG_005905.2:60000-70000	data/brca1_10kb.fa; data/crispr_fresh_pairs.tsv
7	CRISPOR reference CFD/MIT matrices	crisporWebsite (PMID 30341956)	data/crispr_cfd_oracle.tsv; sweep/_crispor_mit_ref.py
8	CRISPOR Rule Set 1 reference	crisporEffScores.py	data/doench_rs1_oracle.tsv
9	UniProtKB reviewed globins	25 sequences	data/globins25.fa
10	UniProtKB reviewed non-globins	25 sequences	data/nonglobins25.fa
11	Pfam Globin HMM	PF00042.29	data/pfam_PF00042_globin.hmm
12	Rfam tRNA seed	RF00005	data/rfam/RF00005.sto
13	Rfam 5S rRNA seed	RF00001	data/rfam/RF00001.sto
14	Rfam RNase P RNA seed	RF00010	data/rfam/RF00010.sto
15	ChEMBL approved drugs	1000-compound slice	data/chembl_approved_1000.tsv
16	Kazusa CUTG E. coli K12	taxid 83333	data/README.md (sanity only)
17	NCBI RefSeq E. coli K-12 MG1655 CDS	GCF.000005845.2	sweep/codon_cai_bench.py
18	PaxDb E. coli integrated abundance	511145-WHOLE_ORGANISM-integrated	sweep/codon_cai_bench.py
19	CPIC CYP2C19 diplotype table	666 diplotypes	sweep/pgx_cpic_bench.py
20	CPIC CYP2D6 diplotype table	16836 diplotypes	sweep/pgx_cpic_bench.py
21	CPIC CYP2D6 allele activity table	208 alleles	sugarcodes-ai pgx_guidelines/data
22	Edinburgh Genome Foundry codon tables	e.coli_316407	sweep/codon_cai_bench.py (via sugarcodes-ai)

C Accession-Level Dataset Manifest

Each row is one identifier-backed record fetched individually and used in a committed analysis (counting rule in manifests/README.md).

#	Kind	Study	Accession	Used in
1	study	1	variant_summary.txt.gz (GRCh38)	data/clinvar_sample_500.tsv; discovery/splice_region_vus/
2	study	2	hg38.2bit	discovery/splice_region_vus/extract.py
3	study	3	top-20 VUS query	discovery/splice_region_vus/lovd_top20.json
4	study	4	skempi_v2.csv (PMID 30598530)	data/skempi_v2.csv
5	study	5	e_coli_core (PMID 21988831)	data/e_coli_core.json
6	study	6	NG_005905.2:60000-70000	data/brca1_10kb.fa; data/crispr_fresh_pairs.tsv
7	study	7	crisporWebsite (PMID 30341956)	data/crispr_cfd_oracle.tsv; sweep/crispor_mit_ref.py
8	study	8	crisporEffScores.py	data/doench_rs1_oracle.tsv
9	study	9	25 sequences	data/globins25.fa
10	study	10	25 sequences	data/nonglobins25.fa
11	study	11	PF00042.29	data/pfam.PF00042_globin.hmm
12	study	12	RF00005	data/rfam/RF00005.sto
13	study	13	RF00001	data/rfam/RF00001.sto
14	study	14	RF00010	data/rfam/RF00010.sto
15	study	15	1000-compound slice	data/chembl_approved_1000.tsv
16	study	16	taxid 83333	data/README.md (sanity only)
17	study	17	GCF_000005845.2	sweep/codon_cai_bench.py
18	study	18	511145- WHOLE_ORGANISM- integrated	sweep/codon_cai_bench.py
19	study	19	666 diplotypes	sweep/pgx_cplic_bench.py
20	study	20	16836 diplotypes	sweep/pgx_cplic_bench.py
21	study	21	208 alleles	sugarcode-ai pgx_guidelines/data
22	study	22	e_coli_316407	sweep/codon_cai_bench.py (via sugarcode-ai)
23	study	23	bdata.20130222.mhci.txt	sweep/neohunter_iedb_bench.py
24	study	24	LP_PDBBind.csv	sweep/docking_bench.py
25	pdb_entry		1CNW	sweep/docking_bench.py
26	pdb_entry		1EJN	sweep/docking_bench.py
27	pdb_entry		1F5L	sweep/docking_bench.py
28	pdb_entry		1G54	sweep/docking_bench.py
29	pdb_entry		1GI1	sweep/docking_bench.py
30	pdb_entry		1K1J	sweep/docking_bench.py
31	pdb_entry		1O3L	sweep/docking_bench.py
32	pdb_entry		1P1O	sweep/docking_bench.py
33	pdb_entry		1Y3X	sweep/docking_bench.py
34	pdb_entry		1ZGI	sweep/docking_bench.py
35	pdb_entry		2ERZ	sweep/docking_bench.py
36	pdb_entry		2J4I	sweep/docking_bench.py
37	pdb_entry		2P4J	sweep/docking_bench.py
38	pdb_entry		2X7T	sweep/docking_bench.py
39	pdb_entry		2XAB	sweep/docking_bench.py
40	pdb_entry		2XJG	sweep/docking_bench.py
41	pdb_entry		2ZDN	sweep/docking_bench.py
42	pdb_entry		3B3C	sweep/docking_bench.py
43	pdb_entry		3C84	sweep/docking_bench.py
44	pdb_entry		3D8W	sweep/docking_bench.py
45	pdb_entry		3HKQ	sweep/docking_bench.py
46	pdb_entry		3HLL	sweep/docking_bench.py
47	pdb_entry		3IBI	sweep/docking_bench.py
48	pdb_entry		3M35	sweep/docking_bench.py
49	pdb_entry		3PO6	sweep/docking_bench.py
50	pdb_entry		3R17	sweep/docking_bench.py
51	pdb_entry		3R4N	sweep/docking_bench.py
52	pdb_entry		4B2I	sweep/docking_bench.py
53	pdb_entry		4DB7	sweep/docking_bench.py

#	Kind	Study	Accession	Used in
54	pdb_entry		4F5Y	sweep/docking_bench.py
55	pdb_entry		4FL1	sweep/docking_bench.py
56	pdb_entry		4J93	sweep/docking_bench.py
57	pdb_entry		4JFM	sweep/docking_bench.py
58	pdb_entry		4KNJ	sweep/docking_bench.py
59	pdb_entry		4LXZ	sweep/docking_bench.py
60	pdb_entry		4MO8	sweep/docking_bench.py
61	pdb_entry		4NH7	sweep/docking_bench.py
62	pdb_entry		4O04	sweep/docking_bench.py
63	pdb_entry		4OAG	sweep/docking_bench.py
64	pdb_entry		4R76	sweep/docking_bench.py
65	pdb_entry		4RFC	sweep/docking_bench.py
66	pdb_entry		4TIM	sweep/docking_bench.py
67	pdb_entry		4XIT	sweep/docking_bench.py
68	pdb_entry		4YTC	sweep/docking_bench.py
69	pdb_entry		4YXI	sweep/docking_bench.py
70	pdb_entry		4ZME	sweep/docking_bench.py
71	pdb_entry		5AML	sweep/docking_bench.py
72	pdb_entry		5CAP	sweep/docking_bench.py
73	pdb_entry		5CAU	sweep/docking_bench.py
74	pdb_entry		5D3L	sweep/docking_bench.py
75	pdb_entry		5EWK	sweep/docking_bench.py
76	pdb_entry		5EWY	sweep/docking_bench.py
77	pdb_entry		5F60	sweep/docking_bench.py
78	pdb_entry		5F62	sweep/docking_bench.py
79	pdb_entry		5G5V	sweep/docking_bench.py
80	pdb_entry		5HTZ	sweep/docking_bench.py
81	pdb_entry		5IX0	sweep/docking_bench.py
82	pdb_entry		5L8C	sweep/docking_bench.py
83	pdb_entry		5MJN	sweep/docking_bench.py
84	pdb_entry		5MPN	sweep/docking_bench.py
85	pdb_entry		5N24	sweep/docking_bench.py
86	pdb_entry		5O87	sweep/docking_bench.py
87	pdb_entry		5SZ4	sweep/docking_bench.py
88	pdb_entry		5SZ5	sweep/docking_bench.py
89	pdb_entry		5Y8Y	sweep/docking_bench.py
90	pdb_entry		6EQ5	sweep/docking_bench.py
91	pdb_entry		6G38	sweep/docking_bench.py
92	pdb_entry		6G6T	sweep/docking_bench.py
93	pdb_entry		6HQY	sweep/docking_bench.py
94	pdb_entry		6HRQ	sweep/docking_bench.py
95	pdb_entry		6OE0	sweep/docking_bench.py
96	pdb_entry		6RFN	sweep/docking_bench.py
97	pdb_entry		6U6W	sweep/docking_bench.py
98	species		10090-ASTROCYTE-integrated.txt	sweep/codon_cai_multispecies.py
99	species		GCF_000001635.27	sweep/codon_cai_multispecies.py
100	species		10116-CARDIAC_VENTRICLE-integrated.txt	sweep/codon_cai_multispecies.py
101	species		GCF_036323735.1	sweep/codon_cai_multispecies.py
102	species		121845-WHOLE_ORGANISM-integrated.txt	sweep/codon_cai_multispecies.py
103	species		GCF_000475195.1	sweep/codon_cai_multispecies.py
104	species		1295-WHOLE_ORGANISM-integrated.txt	sweep/codon_cai_multispecies.py
105	species		GCF_025558325.1	sweep/codon_cai_multispecies.py
106	species		1314-WHOLE_ORGANISM-integrated.txt	sweep/codon_cai_multispecies.py
107	species		GCF_900475035.1	sweep/codon_cai_multispecies.py
108	species		185431-WHOLE_ORGANISM-integrated.txt	sweep/codon_cai_multispecies.py
109	species		GCF_000002445.2	sweep/codon_cai_multispecies.py

#	Kind	Study	Accession	Used in
110	species		192222- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
111	species		GCF_000009085.1	sweep/codon_cai_multispecies.py
112	species		196627- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
113	species		GCF_000011325.1	sweep/codon_cai_multispecies.py
114	species		198214- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
115	species		GCF_000006925.2	sweep/codon_cai_multispecies.py
116	species		208964- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
117	species		GCF_000006765.1	sweep/codon_cai_multispecies.py
118	species		224308- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
119	species		GCF_000009045.1	sweep/codon_cai_multispecies.py
120	species		224911- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
121	species		GCF_001642675.1	sweep/codon_cai_multispecies.py
122	species		243159- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
123	species		GCF_000021485.1	sweep/codon_cai_multispecies.py
124	species		243230- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
125	species		GCF_020546685.1	sweep/codon_cai_multispecies.py
126	species		246200- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
127	species		GCF_000011965.2	sweep/codon_cai_multispecies.py
128	species		2903- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
129	species		GCF_000372725.1	sweep/codon_cai_multispecies.py
130	species		343509- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
131	species		GCF_000010085.1	sweep/codon_cai_multispecies.py
132	species		36329- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
133	species		GCF_000002765.6	sweep/codon_cai_multispecies.py
134	species		403677- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
135	species		GCF_000142945.1	sweep/codon_cai_multispecies.py
136	species		4081-PERICARP- integrated.txt	sweep/codon_cai_multispecies.py
137	species		GCF_036512215.1	sweep/codon_cai_multispecies.py
138	species		4097-LEAF-integrated.txt	sweep/codon_cai_multispecies.py
139	species		GCF_000715075.1	sweep/codon_cai_multispecies.py
140	species		418985- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
141	species		GCA_002081605.1	sweep/codon_cai_multispecies.py
142	species		44689- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
143	species		GCF_000004695.1	sweep/codon_cai_multispecies.py

#	Kind	Study	Accession	Used in
144	species		4932- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
145	species		GCF_000146045.2	sweep/codon_cai_multispecies.py
146	species		54291- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
147	species		GCF_030505655.1	sweep/codon_cai_multispecies.py
148	species		6239- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
149	species		GCF_000002985.6	sweep/codon_cai_multispecies.py
150	species		7227-SPERM-integrated.txt	sweep/codon_cai_multispecies.py
151	species		GCF_000001215.4	sweep/codon_cai_multispecies.py
152	species		7460-ABDOMEN- integrated.txt	sweep/codon_cai_multispecies.py
153	species		GCF_003254395.2	sweep/codon_cai_multispecies.py
154	species		8030-LIVER-integrated.txt	sweep/codon_cai_multispecies.py
155	species		GCF_965601325.2	sweep/codon_cai_multispecies.py
156	species		83332- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
157	species		GCF_000195955.2	sweep/codon_cai_multispecies.py
158	species		9031-HEART-integrated.txt	sweep/codon_cai_multispecies.py
159	species		GCF_016699485.2	sweep/codon_cai_multispecies.py
160	species		93061- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
161	species		GCF_000013425.1	sweep/codon_cai_multispecies.py
162	species		9544-BILE_DUCT- integrated.txt	sweep/codon_cai_multispecies.py
163	species		GCF_049350105.2	sweep/codon_cai_multispecies.py
164	species		9606-ADRENAL_GLAND- integrated.txt	sweep/codon_cai_multispecies.py
165	species		GCF_000001405.40	sweep/codon_cai_multispecies.py
166	species		9615-PLASMA- integrated.txt	sweep/codon_cai_multispecies.py
167	species		GCF_011100685.1	sweep/codon_cai_multispecies.py
168	species		9823-LIVER-integrated.txt	sweep/codon_cai_multispecies.py
169	species		GCF_054392235.1	sweep/codon_cai_multispecies.py
170	species		9913- LENS_OF_CAMERA_TYPE_EYE- integrated.txt	sweep/codon_cai_multispecies.py
171	species		GCF_002263795.3	sweep/codon_cai_multispecies.py
172	species		99287- WHOLE_ORGANISM- integrated.txt	sweep/codon_cai_multispecies.py
173	species		GCF_000006945.2	sweep/codon_cai_multispecies.py
174	species		9986- PERIPHERAL_REGION_OF_RETINA- integrated.txt	sweep/codon_cai_multispecies.py
175	species		GCF_964237555.1	sweep/codon_cai_multispecies.py
176	gtrnadb		Esch.coli_K_12_MG1655	sweep/codon_tai_enc.py
177	gtrnadb		Baci_subt_subtilis_168	sweep/codon_tai_enc.py
178	gtrnadb		Scere3	sweep/codon_tai_enc.py
179	refseq_transcript	25	NM_000546.6	sweep/primer_vs_primer3.py
180	refseq_transcript	25	NM_007294.4	sweep/primer_vs_primer3.py
181	refseq_transcript	25	NM_000059.4	sweep/primer_vs_primer3.py
182	refseq_transcript	25	NM_004333.6	sweep/primer_vs_primer3.py
183	refseq_transcript	25	NM_005228.5	sweep/primer_vs_primer3.py
184	study	26	clinvar.vcf.gz	sweep/vcf_vs_pysam.py
185	study	27	GSE52778	sweep/de_vs_pydeseq2.py

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